

Osu! vs Geometry Dash

A Statistical Analysis of Dominance in Competitive Domains.

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Abstract

This paper investigates whether statistical tools such as standard deviation and z-scores can be used to compare exceptional performance across sports, esports and other competitive fields. By analyzing selected athletes and players relative to their competitive fields, this study aims to explore whether greatness can be measured by statistical deviation from the average.

1 Introduction

1.1 Hypothesis

The ceiling of how good someone can be at something can be observed by the curve of spreads between the best and the average in any given field.

1.2 Explanation

Accurate skill evaluation systems approximate a normally distributed curve. Thus, at the elite tail of a field, the magnitude of standardized separation decays.

To standardize leaderboards and data for comparison, the data must follow a common relative scale. To achieve this, the data that will be taken (Top 100 of a field) will be compared to the average to turn it into a Z-score.

Picture an approximation of a normal distribution, this investigation omits the bottom $\sim 99.9\%$ of the graph, taking the elite scores and standardizing them to the magnitude of spread.

This setup aims to show the comparison of the decay pattern of the spread in the elite tail of the distribution.

2 Research Question

Main question:

How does elite performance decay across the top 100 players in a particular competitive field?

Sub-topics:

- Whats harder? Geometry Dash or Osu!
- Can different domains be compared fairly using the same statistical methods?
- Which competitive domain has the largest statistical outlier?

3 Statistical Background

A brief list of the statistical methods used to process raw data in this investigation.

3.1 Statistical Vectors

Since each competitor may have multiple statistics, their performance can be represented as a vector. For example, a competitor's standardized statistics can be written as:

$$\vec{z} = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ \vdots \\ z_n \end{bmatrix}$$

where each z_i represents the z-score of one statistic.

A weighted composite score can then be interpreted as a dot product between a weight vector and a standardized statistics vector:

$$S = \vec{w} \cdot \vec{z}$$

where $\vec{w}$ is the weight vector and $\vec{z}$ is the standardized performance vector.

3.2 Mean

For ungrouped data, the mean is calculated as:

$$\mu = \frac{\sum x_i}{n}$$

For data with frequencies, the mean is calculated as:

$$\mu = \frac{\sum f_i x_i}{\sum f_i}$$

3.3 Standard Deviation

Standard deviation measures how spread out the values in a dataset are. A low standard deviation means the values are close to the mean, while a high standard deviation means the values are more spread out.

For population data, standard deviation is calculated as:

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{n}}$$

For sample data, standard deviation is calculated as:

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

For data with frequencies, the population standard deviation is:

$$\sigma = \sqrt{\frac{\sum f_i (x_i - \mu)^2}{\sum f_i}}$$

3.4 Z-Score

A z-score measures how many standard deviations a value is above or below the mean. It is useful for comparing values from different datasets because it standardizes them onto the same scale.

$$z = \frac{x - \mu}{\sigma}$$

where x is the observed value, μ is the mean, and σ is the standard deviation.

4 Methodology

The average professional competitor provides a baseline for the field. The top ten average shows how exceptional the best competitor is among elite competitors. The gap between the first and second ranked competitors measures whether the best competitor is clearly separated from their closest rival.

4.1 One Variable Statistical Calculation

4.2 Two Variable Statistical Calculation

4.3 Comparison Methods

To further compare the similarity of decay in top play of any given field, this investigation will compare the similarity angle of statistical vectors through cosine similarity.

The statistical vectors will compare the following:

$$\alpha = z_{\text{avg gap}} = \frac{S_{\text{best}} - \mu_{\text{field}}}{\sigma_{\text{field}}}$$

where S_{best} is the best competitor's score, μ_{field} is the mean score of the comparison field, and σ_{field} is the standard deviation of the comparison field. This measures how many standard deviations the best competitor is above the average competitor.

$$\beta = z_{\text{elite}} = \frac{S_{\text{best}} - \mu_{\text{top 10}}}{\sigma_{\text{top 10}}}$$

where $\mu_{\text{top 10}}$ is the mean score of the top ten competitors and $\sigma_{\text{top 10}}$ is the standard deviation of the top ten competitors. This measures whether the best competitor is still an outlier among elite competitors.

$$G = S_1 - S_2$$

where S_1 is the highest score and S_2 is the second highest score.

$$\gamma = z_{\text{field gap}} = \frac{G}{\sigma_{\text{field}}}$$

where γ measures how large the first-second gap is relative to the spread of the broader field.

$$\phi = z_{\text{elite gap}} = \frac{G}{\sigma_{\text{top 10}}}$$

where ϕ measures how large the first-second gap is relative to the spread of the top ten competitors.

5 Data Collection